Susceptibility and Spatial Signature of Abrupt Thaw Across Alaska's Permafrost Landscapes

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Key points

- Key point 1
- Key point 2
- Key point 3

Abstract

Abstract text body.

Plain Language Summary

Plain-language summary body.

1 Introduction

The northern permafrost region contains more organic carbon in its upper three meters of soil than currently resides in Earth's atmosphere (Hugelius et al., 2014). Most permafrost thaw proceeds by gradual, top-down deepening of the active layer, but abrupt thaw destabilizes and collapses the ground surface over days to years, mobilizing carbon stocks that gradual thaw would take decades to reach (Turetsky et al., 2020; Nitzbon et al., 2020). Under high emissions scenarios, abrupt thaw is projected to impact only 2.5 million km^2 out of the 18 million km^2 permafrost region, but threatens to release nearly as much carbon as gradual thaw processes operating over a much larger area (Turetsky et al., 2020). Abrupt thaw leads to a variety of landforms, including thermokarst lakes, collapsing peatlands, and hillslope failures, all of which differ in the quantity and form of carbon released (Olefeldt et al., 2016; Schädel et al., 2016). The scale and timing of carbon release from permafrost soils therefore depend on how and where the landscape undergoes abrupt thaw, and this spatial pattern remains one of the largest sources of uncertainty in the permafrost-carbon feedback (Turetsky et al., 2020; Walter Anthony et al., 2018).

Beyond climate feedbacks, abrupt thaw represents an immediate hazard to the roughly 5 million people who live in the Arctic circumpolar permafrost region (Ramage et al., 2021). Projections suggest that 30-50% of Arctic infrastructure may be at risk of damage by permafrost thaw by 2050, threatening tens of billions of U.S. dollars in damage by the end of the century (Hjort et al., 2022). In Alaska alone, permafrost thaw may cause $37 to $51 billion in damage to buildings and roads (Manos et al., 2025). This burden will fall disproportionately on remote and Indigenous communities, whose unofficial, "vernacular" infrastructure and assets often lie outside conventional damage inventories (Verstraete & Zachwieja, 2025). Between 1974 and 2019, thermokarst at Point Lay, Alaska, expanded nearly an order of magnitude faster beneath the developed village than in the surrounding tundra, indicating a potential positive feedback between the built environment and abrupt thaw (Jones et al., 2025). Predicting how and where abrupt thaw will develop on the landscape is therefore critical for helping Arctic communities adapt to or mitigate the effects of future climate change.

Abrupt permafrost thaw proceeds faster than gradual thaw, by definition, but also results from different processes, leaves distinct surface expressions, and has specific consequences for ecosystems, hydrology, and carbon release (Webb et al., 2025). Gradual thaw typically proceeds from the top down as a progressive deepening of the seasonally thawed active layer into near-surface permafrost (Walter Anthony et al., 2018; Webb et al., 2025). Abrupt thaw, by contrast, is not confined to this vertically ordered progression: subsidence, erosion, mass wasting, and the flow of water and heat can rapidly propagate thaw downward and laterally, affecting deeper or adjacent permafrost (Fig. 1) (Turetsky et al., 2020; Nitzbon et al., 2020; Webb et al., 2025). The resulting changes to the landscape occur quickly relative to geomorphic time scales, from vertical subsidence of almost a meter over a decade (Farquharson et al., 2019) to thaw-slump headwall retreat of more than ten meters in a single year (S. Kokelj et al., 2015). A site's thaw mode, abrupt or gradual, is not dictated by the presence of permafrost alone, but emerges from an interplay of ground ice, geomorphology, hydrology, and disturbance history (Jorgenson & Osterkamp, 2005; S. V. Kokelj & Jorgenson, 2013).

Permafrost has been mapped extensively across the Arctic, but existing products largely resolve conditions other than the mode of thaw. Many characterize the substrate, outlining permafrost extent and zonation (Brown et al., 1997), ground ice content (Jorgenson et al., 2008), and probability of occurrence (Obu et al., 2019). Others document where abrupt thaw features have formed, such as ponds and lakes (Muster et al., 2017) or retrogressive thaw slumps (Nitze et al., 2025), or quantify how far degradation of specific features has advanced (Zhang et al., 2025). A growing set of susceptibility maps predicts where specific abrupt thaw processes may develop, including thermokarst lakes (Wang et al., 2023), retrogressive thaw slumps (Makopoulou et al., 2024), and active layer detachment (Makopoulou et al., 2025), and a recent hemispheric model combines lake and slump susceptibility into a single product (Yang et al., 2025). Each of these efforts estimates the likelihood that a named landform will occur at any given point, and the combined model remains a union of predictions anchored to specific processes. None of these previous efforts addresses the question of thaw mode: whether the conditions at a site resemble those under which abrupt or gradual thaw has been observed, independent of any single landform (Fig. 1). Across Alaska, that contrast is carried only by point-wise samples from expert analysis of field and remote sensing observations (Webb et al., 2026), and not yet mapped as a continuous, predictive surface.

Here, we map thaw mode as a continuous, predictive surface across Alaska. We train a gradient-boosted classifier on 19,288 expert-labeled thaw points from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The classifier evaluates sites along a gradient of how strongly key environmental variables resemble either abrupt or non-abrupt thaw, using 70 geospatial indicators sourced from remote sensing, community data products, and model reanalysis. To mitigate sampling bias inherent to the Thaw Database, we evaluate the model under spatial cross-validation, bounded by an applicability mask that indicates where points fall within the envelope of the training data, and then apply it to a statewide feature grid populated by the same data sources. The resulting product is a prior-free, log-evidence index of thaw mode. We then use grouped Shapley (SHAP) values (Lundberg & Lee, 2017; Lundberg et al., 2020) to identify which geospatial indicators best distinguish between thaw modes and how they shift the classifier's prediction. This paper describes the expert-labeled points and feature stack in Section 2, model construction and evaluation in Section 3, the susceptibility map and its drivers in Section 4, and limitations, interpretations, and comparisons with the broader permafrost mapping literature in Section 5.

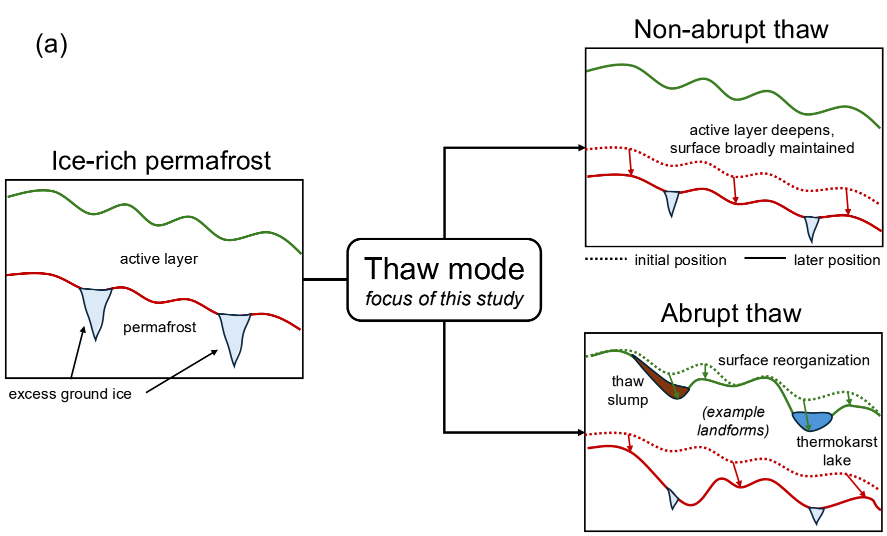


Figure 1.

2 Data

2.1 Labeled thaw points

We train and evaluate the classifier on expert-labeled observations of thaw mode from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The Thaw Database compiles observations of permafrost across Alaska from a wide variety of sources, including both field and remote-sensing studies. After removing exact duplicates in feature space, the training set comprises 19,288 points, each assigned a binary thaw mode label. We encode abrupt thaw as `class 0`, and then all non-abrupt thaw labels as `class 1`. Abrupt thaw dominates the dataset with 18,181 points (94.26%), alongside only 1,107 (5.74%) non-abrupt thaw points. Notably, this composition is the inverse of the landscape it samples. The majority of observations in the Thaw Database are `class 1`, yet abrupt thaw is projected to impact only a fraction of the permafrost region overall (Turetsky et al., 2020).

The non-abrupt thaw label denotes sites undergoing gradual rather than abrupt thaw. Roughly 18% of these points are drawn from the long-term permafrost monitoring networks (the Global Terrestrial Network for Permafrost boreholes and Circumpolar Active Layer Monitoring program). We label these points as "non-abrupt" rather than "gradual" thaw deliberately, following the convention in the Thaw Database, to include sites where thaw is not definitively abrupt. The abrupt thaw class contains a variety of landforms, including thermokarst lakes, collapsing peatlands, retrogressive thaw slumps, and active-layer detachments, among others. We refer the reader to Webb et al. (2026) for the full decision matrix used to label points.

Because the observations were largely assembled from field campaigns and targeted remote-sensing surveys, the spatial distribution of labeled points reflects the research focus of the contributing studies and physical accessibility of sites. Figure 2 shows the distribution of abrupt (a) and non-abrupt (b) thaw observations. The two classes are not evenly balanced across the landscape, with many more abrupt thaw studies focused in the north, and non-abrupt sites spread more evenly across the state. This uneven coverage motivates the spatial cross-validation and area-of-applicability analyses described in Section 3. We carry site coordinates through the pipeline as metadata and use them to build cross-validation blocks, but exclude them from the feature table so that the model cannot use location itself as a predictor.

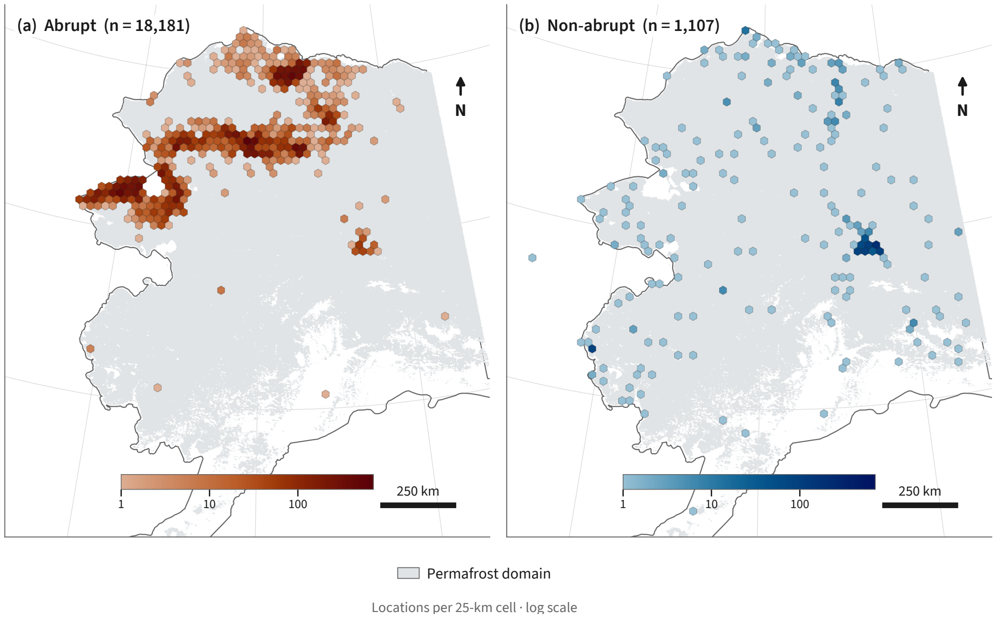


Figure 2.

2.2 Geospatial features

Each point is characterized by a common set of geospatial features extracted at its location, which together form a feature table of 19,288 rows and 70 columns. Continuous variables enter the matrix directly, while categorical variables are one-hot encoded into individual columns. Data sources include remote sensing products, modeled or interpolated data products, and gridded climate products (Table A1).

We obtain surface topography from the USGS 3D Elevation Program 10 m digital elevation model (U.S. Geological Survey, 2019). We use elevation, slope, aspect (decomposed into northness and eastness), and mean curvature computed at 500 m and 2 km scales. We account for surface hydrology using MERIT Hydro (Yamazaki et al., 2019), pulling height above nearest drainage and upstream contributing area. Land cover classifications come from the National Land Cover Database for Alaska (Dewitz, 2019), and are one-hot encoded into 18 indicator columns. Fire history is derived from the MODIS MCD64A1 burned-area product (Giglio et al., 2018), expressed as time since last fire and cumulative burn count.

We obtain soil properties from SoilGrids (Poggio et al., 2021), including organic carbon, total nitrogen, bulk density, and soil texture in sand, silt, and clay fractions. To avoid collinear columns (where sand, silt, and clay fraction sum to a fixed total), we encode soil texture as sand fraction and clay fraction, dropping silt from the feature table. We then aggregate the soil data into two depths, "surface" from 0-30 cm and "depth" from 30-200 cm, to aid in later interpretability. We include relative flammability as a continuous column and dominant vegetation mode encoded into 7 indicator columns, both sourced from the ALFRESCO landscape succession model (Bennett et al., 2017). We include a binary classification of yedoma from the Ice-Rich Yedoma Permafrost Inventory (Strauss et al., 2022), denoting points that are classified as "confirmed" yedoma in the IRYP inventory. Nineteen bioclimatic variables describe annual and seasonal temperature and precipitation, sourced from the WorldClim v1 climatology (Hijmans et al., 2005). Mean annual snow water equivalent (SWE) and multidecadal trends in SWE, precipitation, and temperature come from Daymet v4 (Thornton et al., 2022), computed from 1991-2020.

Across all data products, we do not impute missing values, choosing instead to select a classification algorithm that handles NaN values natively (Section 3). Two sources have structured gaps worth noting: SoilGrids is missing at approximately 16% of our training points, and the MODIS fire domain thins considerably above 70°N, leaving the northern Arctic coast without fire history coverage. Table A1 lists all data products with their native resolutions and references.

2.3 Prediction datacube

Once we have a trained classifier, producing predictions of thaw mode across Alaska requires mapping the same 70 geospatial indicators onto a statewide grid. We build the prediction datacube at a nominal resolution of 1 km on a fixed 0.00898° grid in geographic coordinates (EPSG:4326). True cell area varies with latitude, narrowing towards the north. We sample each data product onto the grid with the exact same extraction and post-processing used for the feature table, described above. Continuous rasters are resampled from their native resolution to the grid resolution, categorical layers are sampled by nearest neighbor, and yedoma is assigned by polygon membership. This parity ensures that the trained model transfers from the training points to the statewide grid without a train/serve mismatch. We restrict all predictions to the permafrost domain defined by the Obu et al. (2019) permafrost probability map, where the classification of thaw mode is physically meaningful.

3 Methods

4 Results

4.1 Abrupt Thaw Susceptibility Map

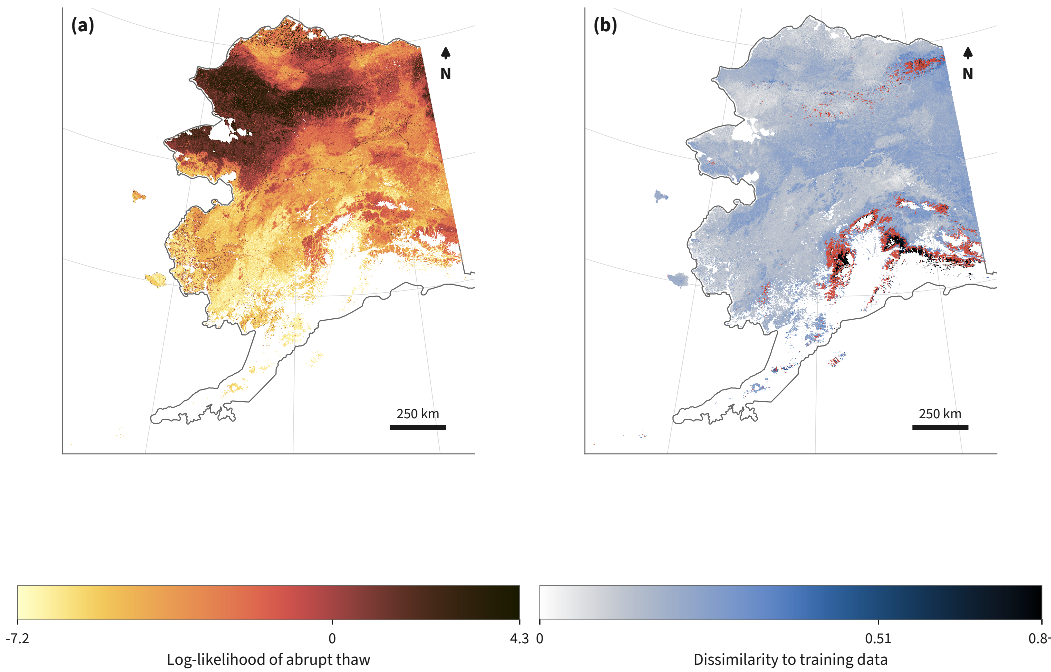


Figure 3.

4.2 Key Indicators of Abrupt Thaw

5 Discussion

6 Conclusion

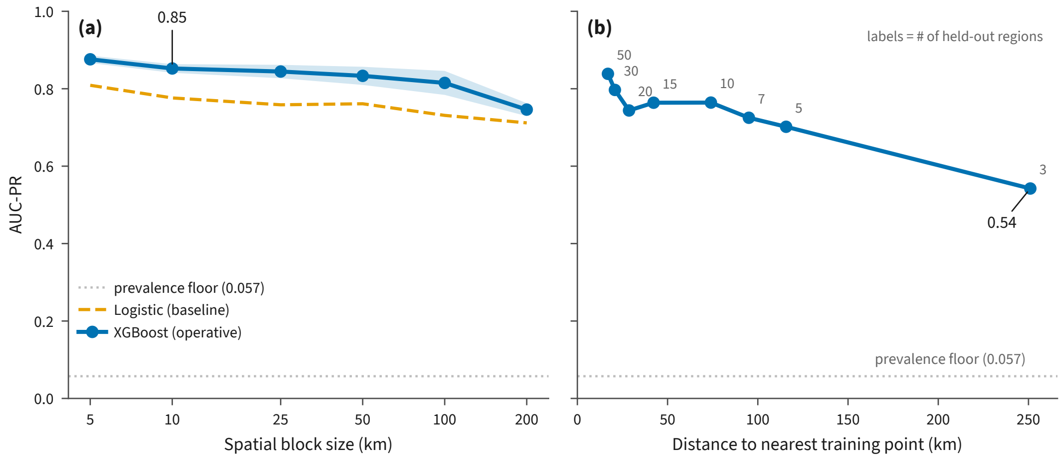


Figure 4.

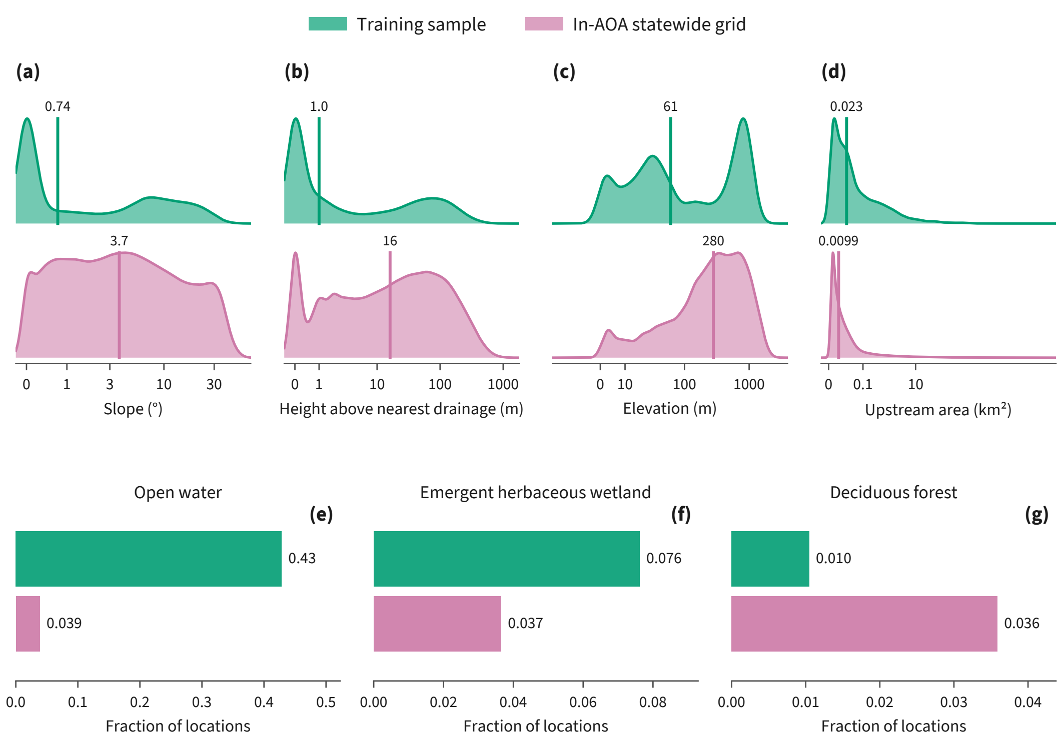


Figure 5.

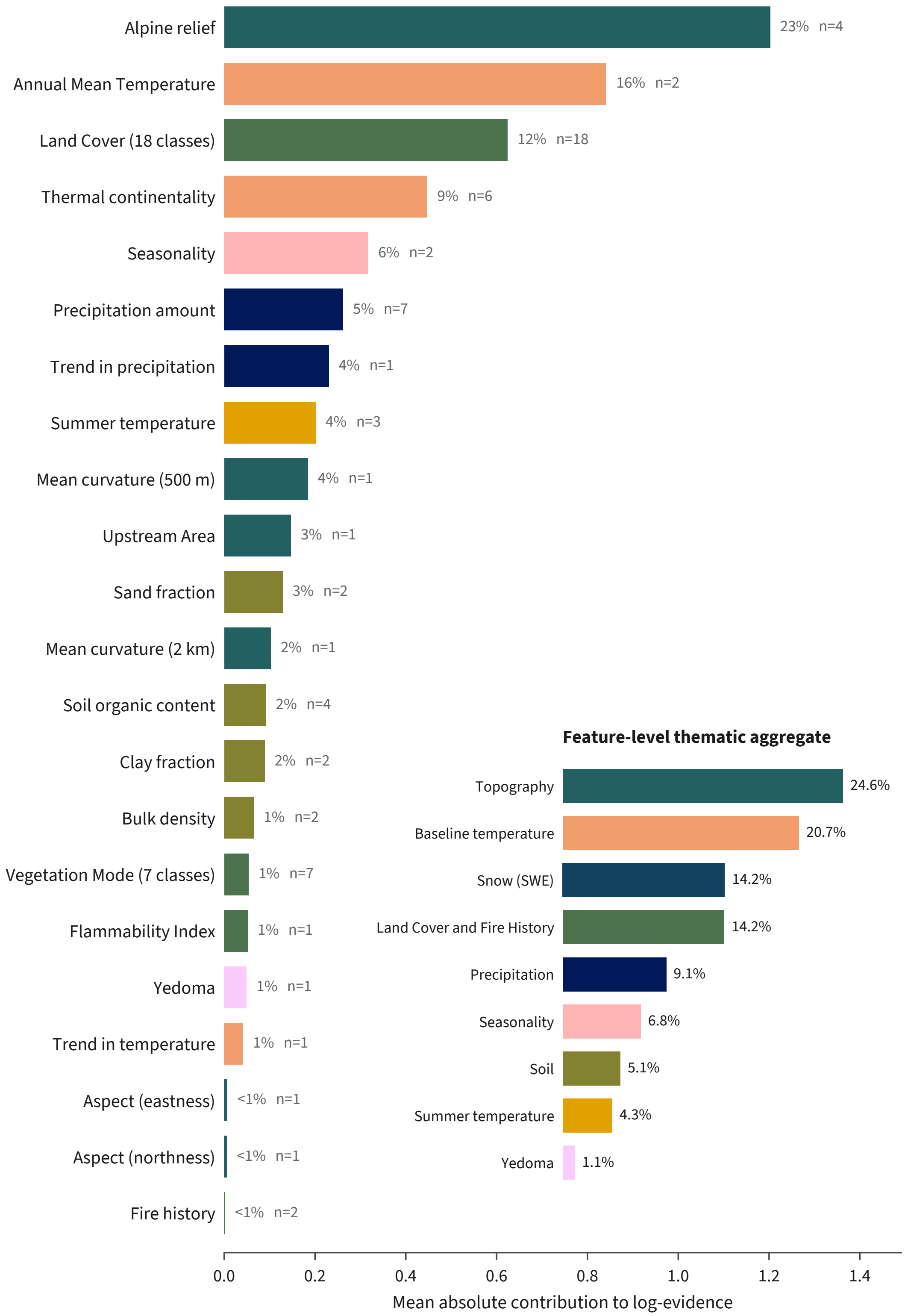

Alpine relief
23% n=4
Annual Mean Temperature
16% n=2
Land Cover (18 classes)
12% n=18
Thermal continentality
9% n=6
Seasonality
6% n=2
Precipitation amount
5% n=7
Trend in precipitation
4% n=1
Summer temperature
4% n=3
Mean curvature (500 m)
4% n=1
Upstream Area
3% n=1
Sand fraction
3% n=2
Mean curvature (2 km)
2% n=1
Soil organic content
2% n=4
Clay fraction
2% n=2
Bulk density
1% n=2
Vegetation Mode (7 classes)
1% n=7
Flammability Index
1% n=1
Yedoma
1% n=1
Trend in temperature
1% n=1
Aspect (eastness)
<1% n=1
Aspect (northness)
<1% n=1
Fire history
<1% n=2
Feature-level thematic aggregate
Topography
24.6%
Baseline temperature
20.7%
Snow (SWE)
14.2%
Land Cover and Fire History
14.2%
Precipitation
9.1%
Seasonality
6.8%
Soil
5.1%
Summer temperature
4.3%
Yedoma
1.1%
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Mean absolute contribution to log-evidence

Figure 6.

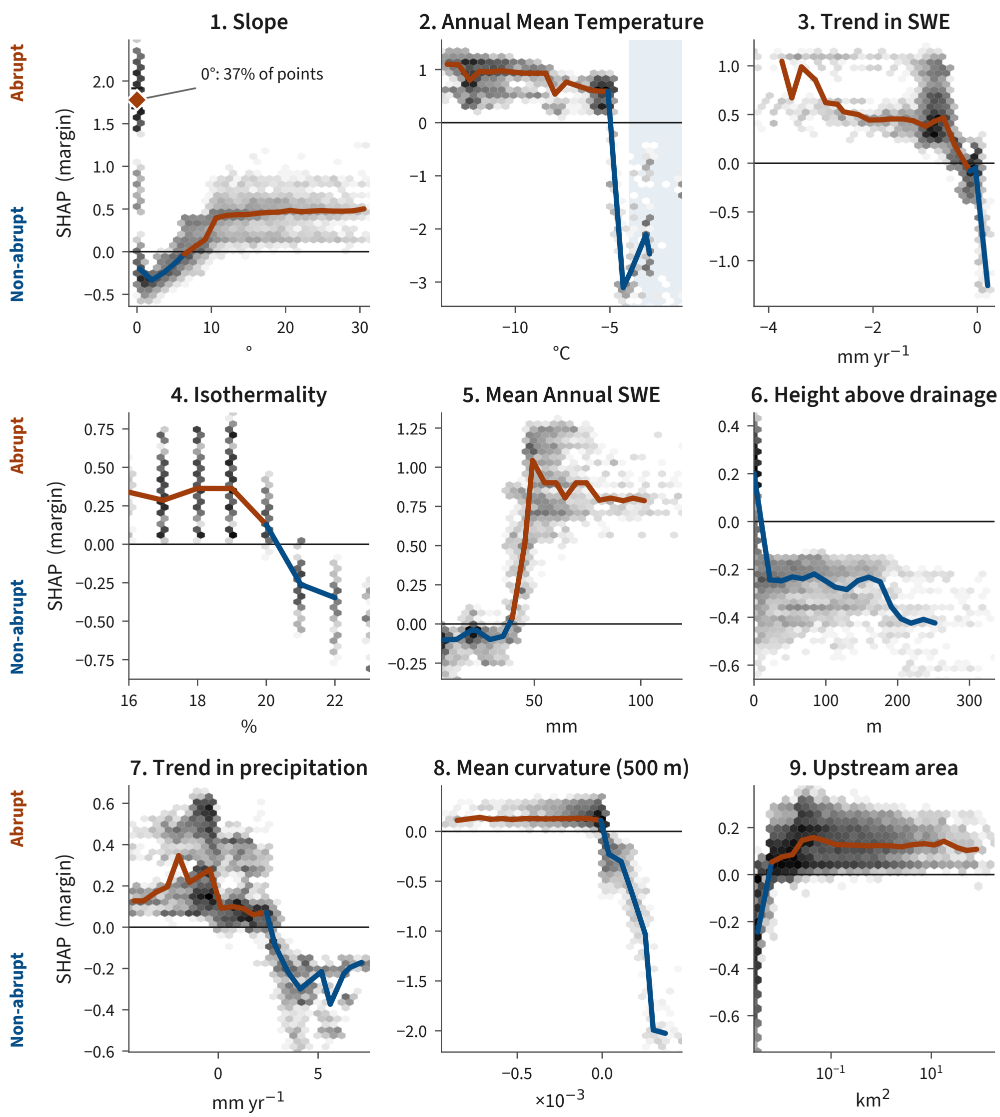

1. Slope
2. Annual Mean Temperature
3. Trend in SWE
4. Isothermality
5. Mean Annual SWE
6. Height above drainage
7. Trend in precipitation
8. Mean curvature (500 m)
9. Upstream area
Abrupt
Non-abrupt
SHAP (margin)
0°: 37% of points
°
°C
mm yr⁻¹
%
mm
m
×10⁻³
km²

Figure 7.

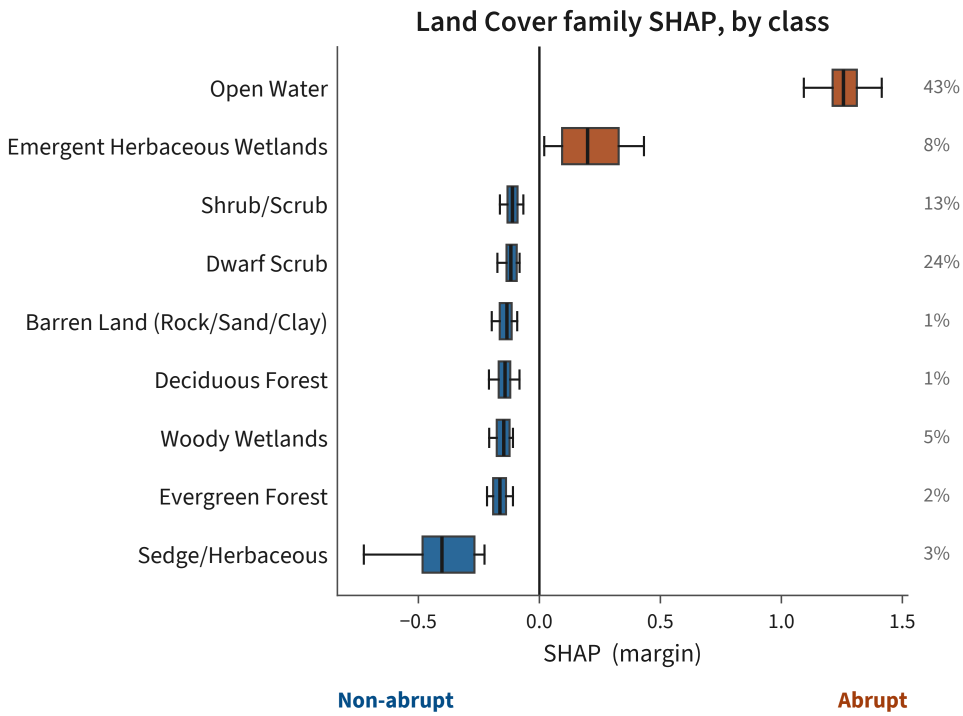


Figure 8.

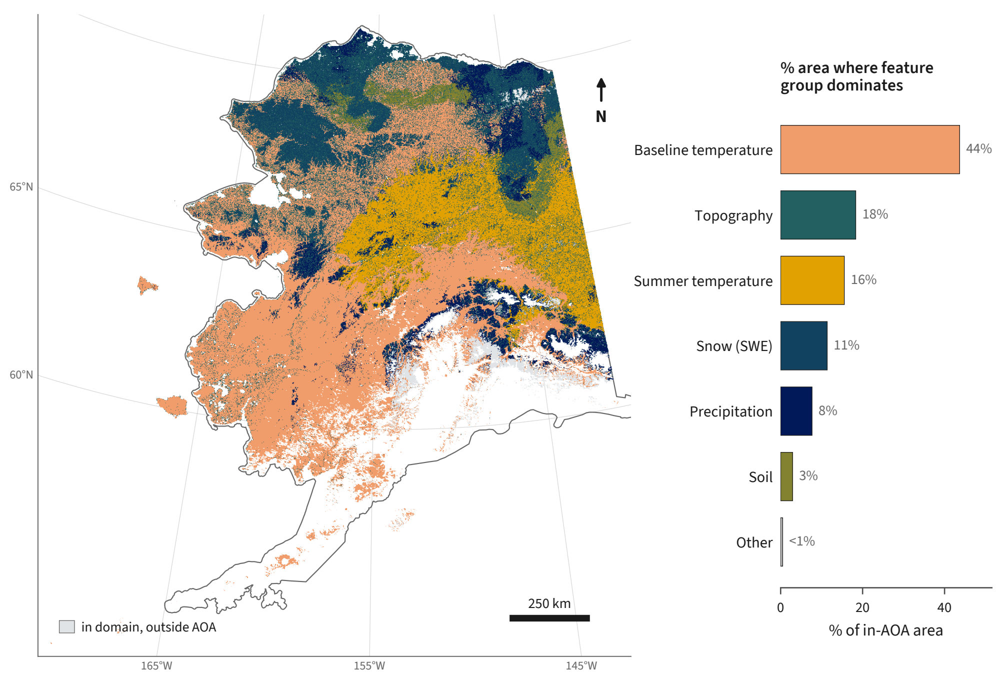


Figure 9.

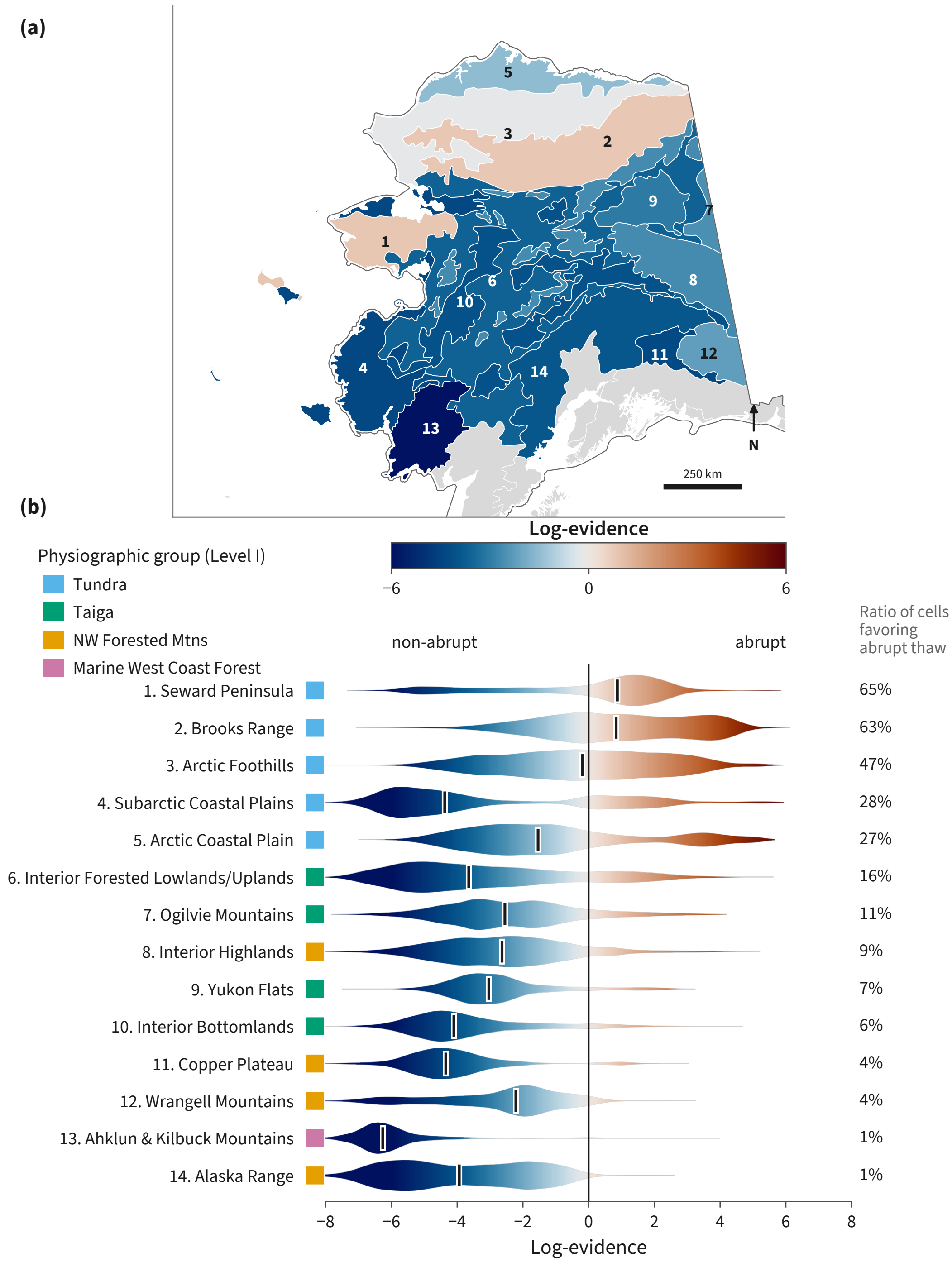
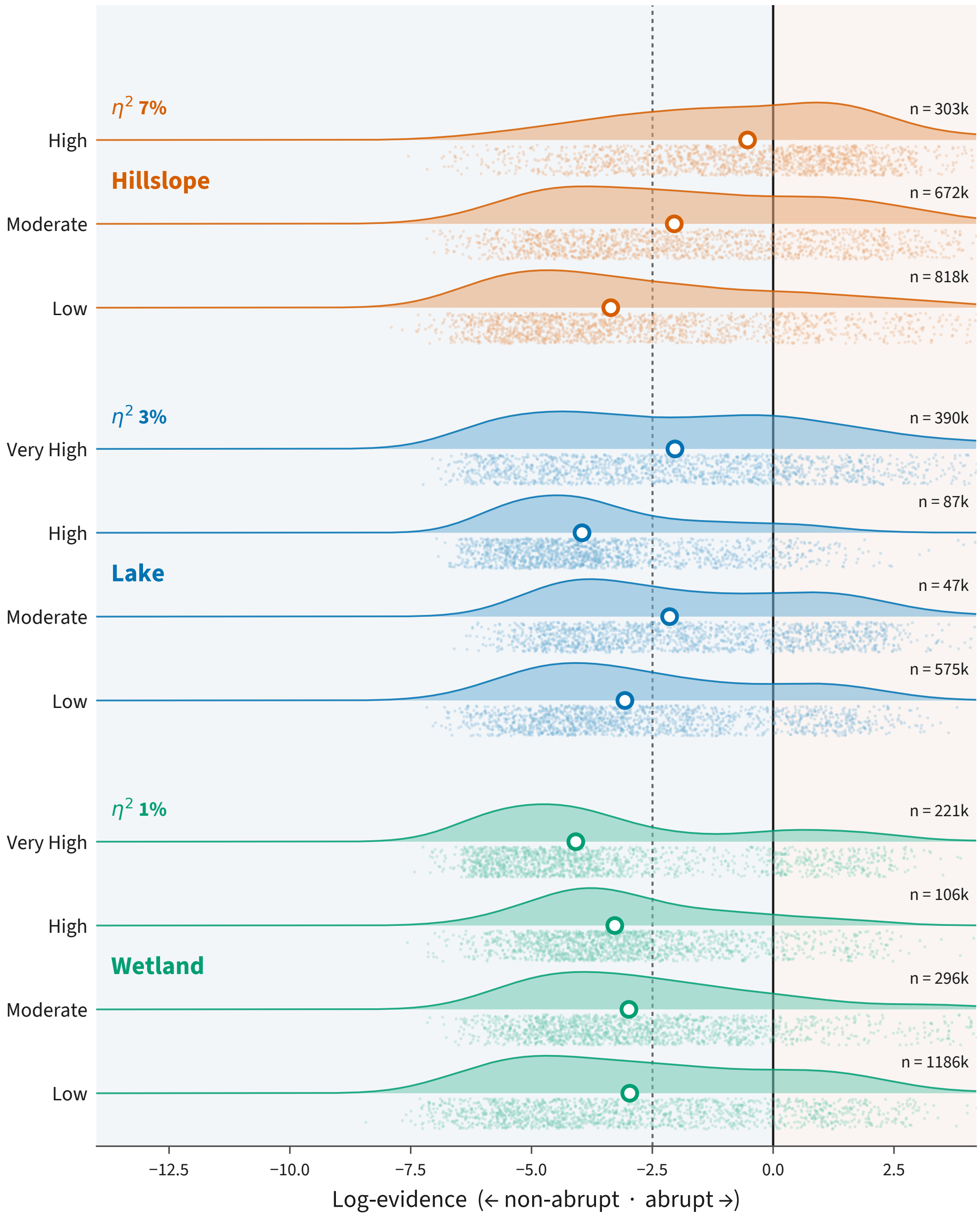

(a)
5
3
2
9
7
1
6
8
10
11
12
4
14
13
N
250 km
(b)
Log-evidence
Physiographic group (Level I)
Tundra
Taiga
NW Forested Mtns
Marine West Coast Forest
−6
0
6
Ratio of cells favoring abrupt thaw
non-abrupt
abrupt
1. Seward Peninsula
2. Brooks Range
3. Arctic Foothills
4. Subarctic Coastal Plains
5. Arctic Coastal Plain
6. Interior Forested Lowlands/Uplands
7. Ogilvie Mountains
8. Interior Highlands
9. Yukon Flats
10. Interior Bottomlands
11. Copper Plateau
12. Wrangell Mountains
13. Ahklun & Kilbuck Mountains
14. Alaska Range
65%
63%
47%
28%
27%
16%
11%
9%
7%
6%
4%
4%
1%
1%
−8
−6
−4
−2
0
2
4
6
8
Log-evidence

Figure 10.



η² 7%
n = 303k
High
Hillslope
n = 672k
Moderate
n = 818k
Low
η² 3%
n = 390k
Very High
n = 87k
High
Lake
n = 47k
Moderate
n = 575k
Low
η² 1%
n = 221k
Very High
n = 106k
High
Wetland
n = 296k
Moderate
n = 1186k
Low
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
Log-evidence (← non-abrupt · abrupt →)

Figure 11.

Open Research Statement

Statement text.

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Appendix A Appendix 1

Table A1. Geospatial data products used to build the feature set. "Cols" gives the number of columns each product contributes to the 70-column model matrix, after one-hot encoding of the categorical land-cover and vegetation-mode sources and depth aggregation of the soil variables.

Data product	Native res.	Variables derived	Cols	Reference
Remote sensing				
USGS 3DEP DEM	10 m	Elevation, slope, aspect (northness, eastness), mean curvature (500 m, 2 km)	6	(U.S. Geological Survey, 2019)
MERIT Hydro v1.0.1	~90 m	Height above nearest drainage, upstream area	2	(Yamazaki et al., 2019)
NLCD 2016 (Alaska)	30 m	Land cover (one-hot)	18	(Dewitz, 2019)
MODIS MCD64A1	~500 m	Time since last fire, burn count	2	(Giglio et al., 2018)
Community and modeled data products				
SoilGrids (ISRIC)	250 m	Organic carbon, nitrogen, bulk density, sand, clay (two depths)	10	(Poggio et al., 2021)
ALFRESCO (SNAP)	~1 km	Flammability, vegetation mode (one-hot)	8	(Bennett et al., 2017)
IRYP v2 (yedoma)	Vector	Confirmed yedoma presence	1	(Strauss et al., 2022)
Gridded climate				
WorldClim v1	~1 km	19 bioclimatic variables	19	(Hijmans et al., 2005)
Daymet V4	1 km	Mean annual SWE, and SWE, precipitation, and temperature trends (1991–2020)	4	(Thornton et al., 2022)
Total			70	